

TTS Model Selection Report

Egyptian Arabic text-to-speech — bake-off results

Best robot realtime

VoiceTut-TTS

Best speed

VoiceTut-TTS

Lowest VRAM

VoiceTut-TTS

OK / attempted

4 / 4

Overview

4 of 4 TTS models generated audio successfully.
Automated scores cover RTF, throughput (chars/s), VRAM, and load — not voice naturalness.
IMPORTANT: listen to each WAV before choosing a production voice.
RTF = generate_time / audio_length. LOWER better. RTF < 1 = realtime-capable.
Chars/s (HIGHER better). Peak VRAM / RAM / load time (LOWER better).
Top robot realtime pick: VoiceTut-TTS (score 90.02).

How to read this report

Page 2: metrics glossary — RTF, chars/s, VRAM, and what to prefer.
Page 3: recommended picks with rationale.
Pages 4-6: robot score, RTF/throughput, and timing charts.
Pages 7-8: VRAM/load and normalized score breakdown.
Page 9: listening checklist and final selection guidance.

TTS - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

Chars/s (characters per second)

HIGHER better

Input characters synthesized per second of compute.
Higher = faster speech generation.

Generation time (s)

LOWER better

Time to synthesize the full utterance. Lower = robot speaks sooner after the LLM.

Audio length (s)

Context only

Duration of the produced WAV. Not good/bad alone — used with generate time to get RTF.

Sec per 1k chars

LOWER better

Seconds of compute per 1000 characters. Lower = more efficient synthesis.

Realtime capable

Prefer YES

Yes when $RTF < 1$ (generate time shorter than the spoken audio).

Peak VRAM (MB)

LOWER better

GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Peak RAM (MB)

LOWER better

System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better

CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better

Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better

Processing time / audio duration. $RTF < 1$ = faster than realtime (good). $RTF > 1$ = slower than the audio.

x Realtime (xRT)

HIGHER better

How many times faster than realtime (about $1/RTF$).
Example: $6x = 6s$ of audio in $\sim 1s$ of compute.

Robot realtime score (0-100)

HIGHER better

Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Balanced score (0-100)

HIGHER better

Normalized blend of the main tradeoffs. Higher = better all-rounder.

Success rate %

HIGHER better

Share of runs that completed without error. Prefer 100% for production reliability.

Remember

For TTS: want LOWER RTF, LOWER generate time, LOWER VRAM/RAM, LOWER load time;
want HIGHER chars/s, HIGHER xRealtime, HIGHER robot score.
Prefer realtime-capable ($RTF < 1$). Always listen to WAVs — scores ignore naturalness.

Recommended picks (automated - listen to confirm)

Best for robot realtime (HIGHER score better)

VoiceTut-TTS

Best composite of RTF + throughput + low VRAM + fast load. Listen: VoiceTut-TTS.wav

score_robot_realtime=90.02 | rtf=0.157 | chars_per_second=82.7 | peak_vram_mb=3150.2

Best speed / lowest RTF (LOWER better)

VoiceTut-TTS

Lowest RTF (generation faster relative to audio length).

rtf=0.157 | x_realtime=6.39 | generation_seconds=20.48

Best throughput chars/s (HIGHER better)

VoiceTut-TTS

Highest characters synthesized per second.

chars_per_second=82.7

Lowest peak VRAM (LOWER better)

VoiceTut-TTS

Lowest peak VRAM — better for 6–16 GB GPUs.

peak_vram_mb=3150.2

Best balanced

VoiceTut-TTS

Balanced speed/throughput/VRAM/generation time.

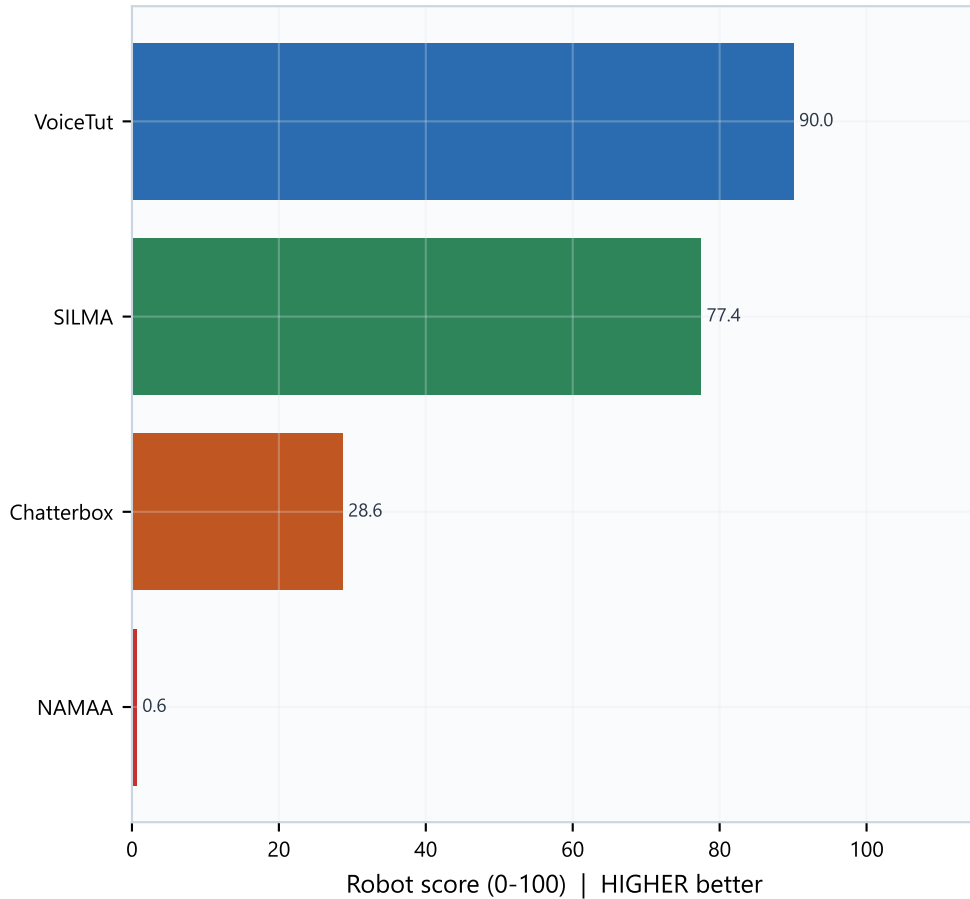
score_balanced=100.0

Selection notes

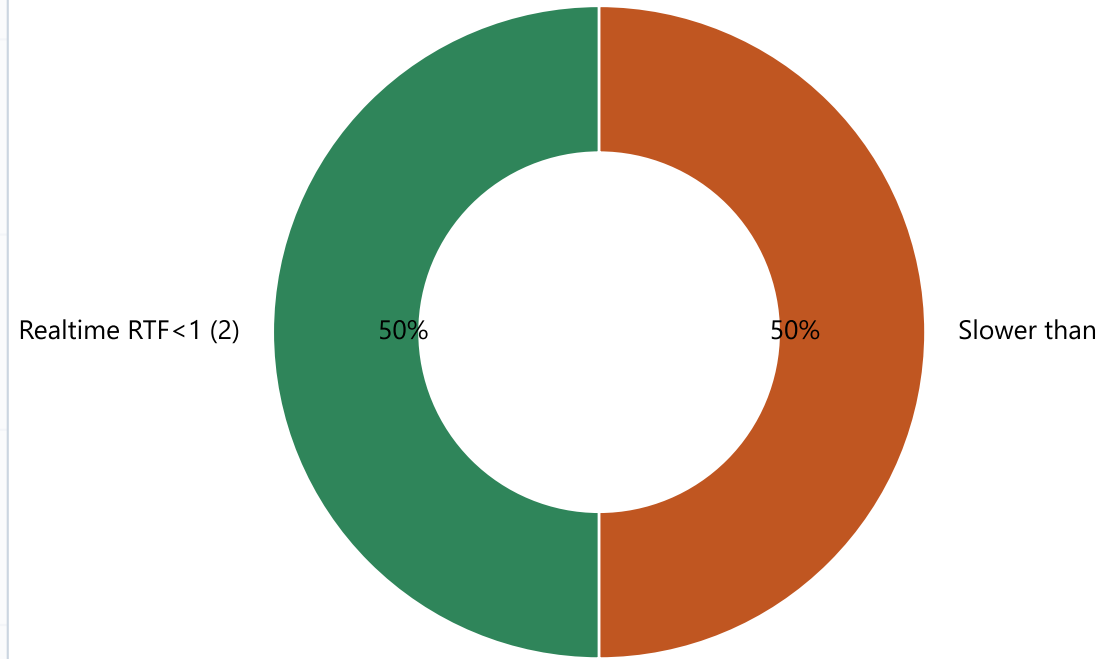
Naturalness / Egyptian dialect / code-switch quality must be judged by listening to WAVs.
Automated scores only cover latency and resource fit for the robot pipeline.
RTF < 1.0 (LOWER better) is preferred for near-real-time synthesis.
For production, also measure time-to-first-audio with streaming APIs.

Robot realtime score & Realtime capability

Composite: RTF + throughput + VRAM + load (HIGHER better)



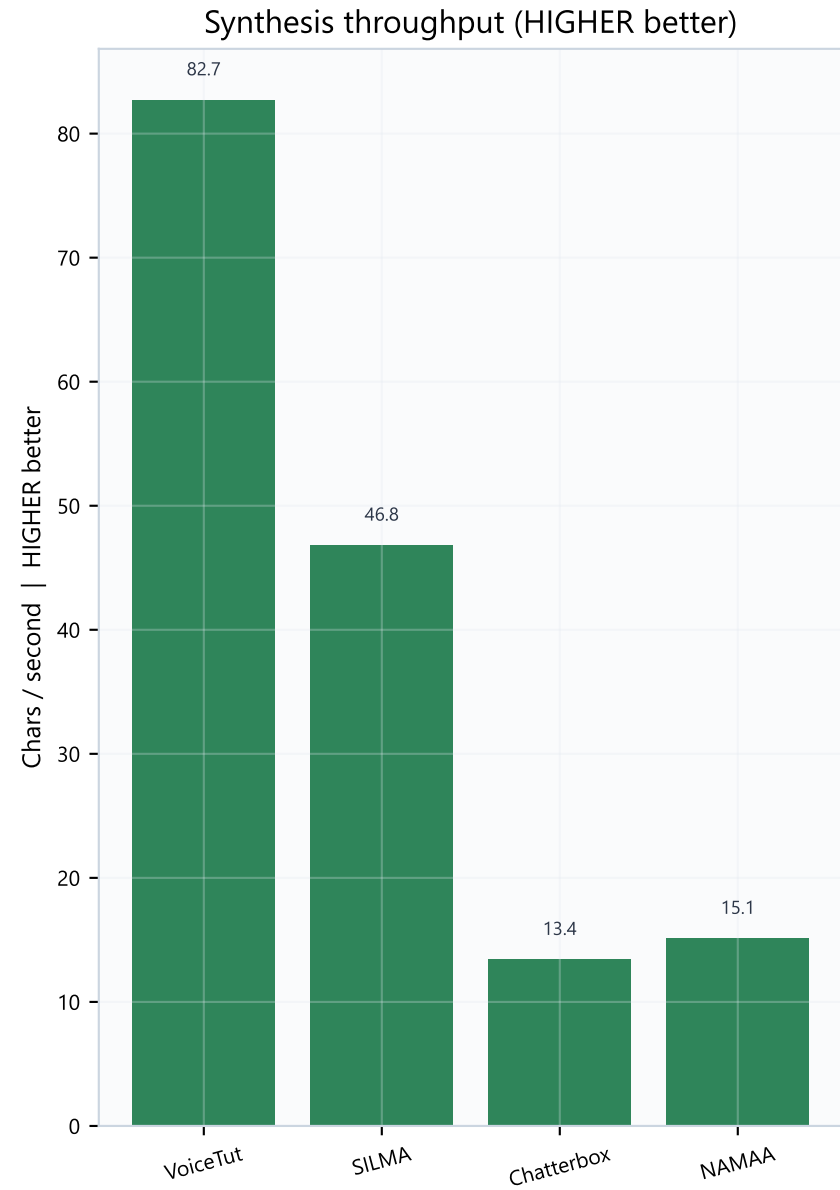
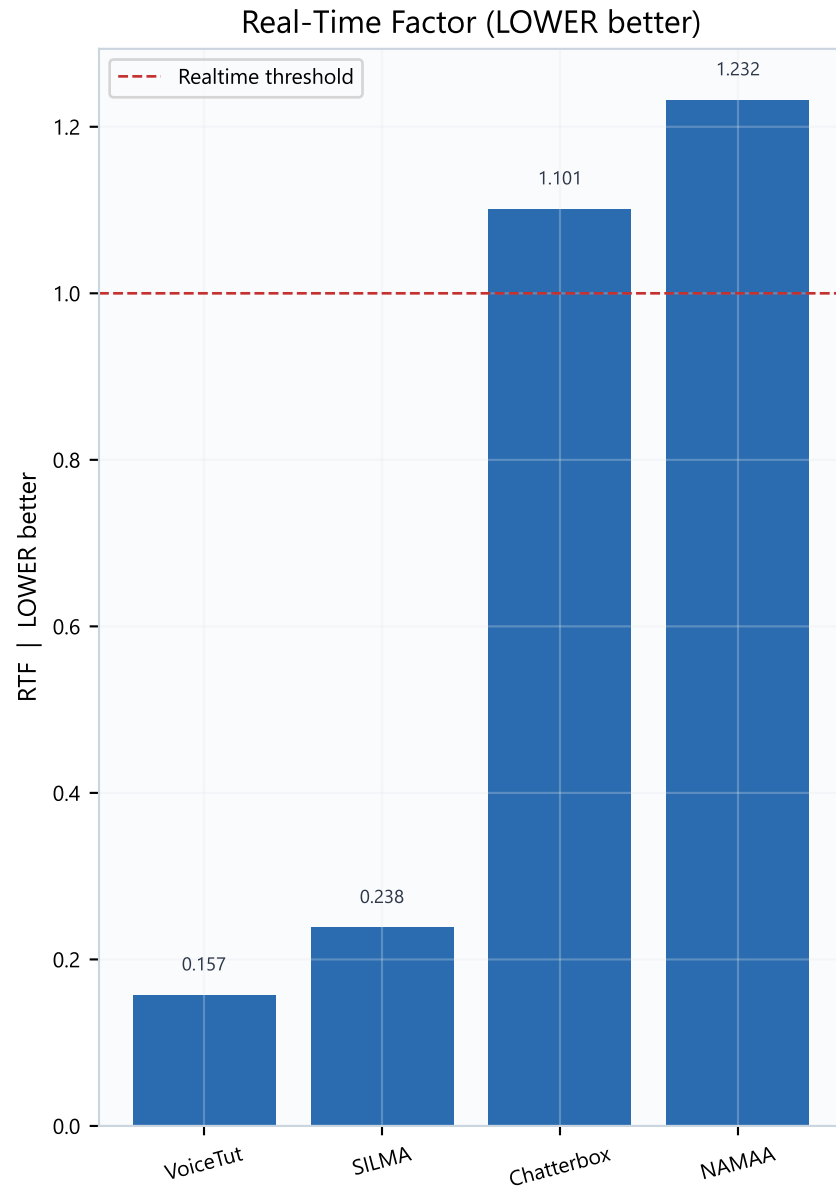
Realtime capable?



What this means

VoiceTut-TTS leads robot realtime with the best blend of speed and low VRAM.
VoiceTut and SILMA are realtime-capable (RTF < 1); Chatterbox and NAMAA are not.
Non-realtime models can still be used offline or for offline rendering, but hurt live robot replies.
Listen before finalizing — a fast voice that sounds unnatural is still a bad production pick.

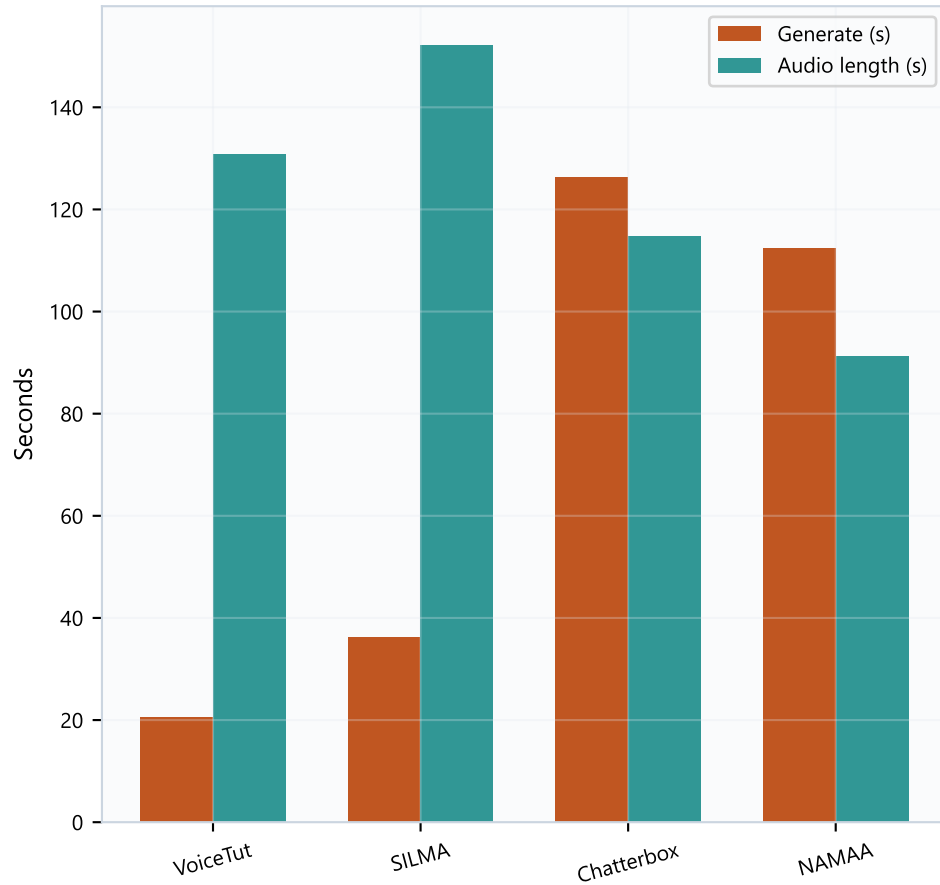
Speed & throughput



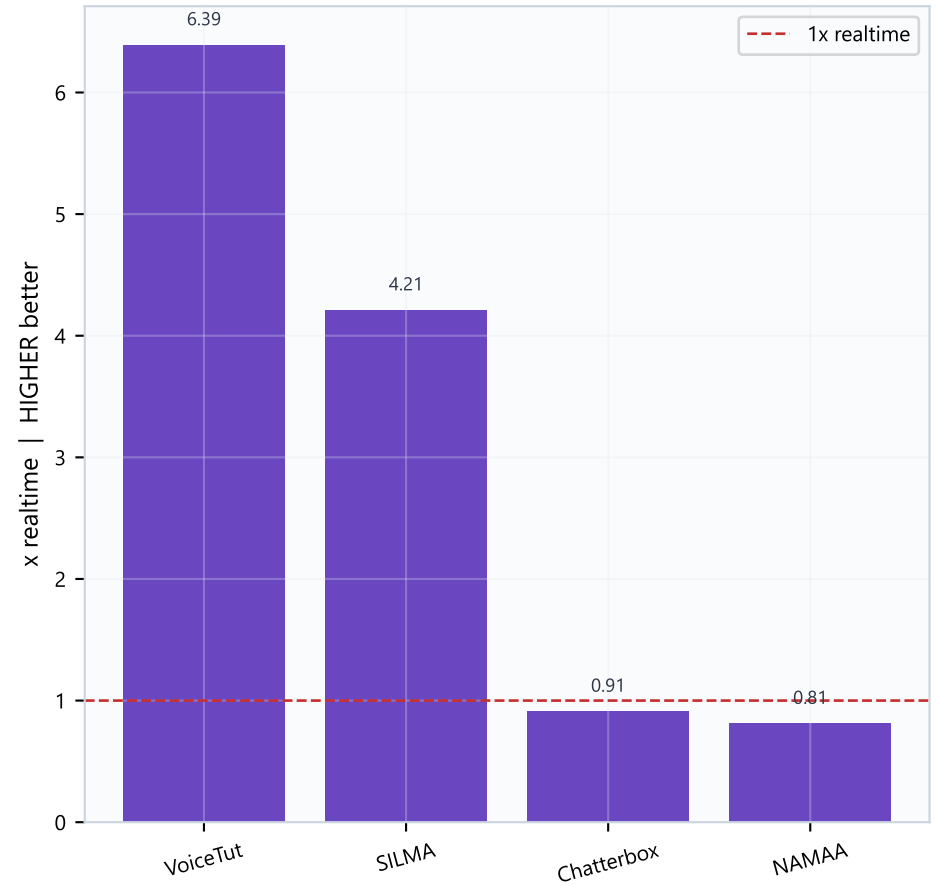
VoiceTut is clearly fastest (RTF 0.157, ~83 chars/s). SILMA is a solid second. Chatterbox and NAMAA take longer than the audio they produce — weak for live robot turns.

Generation time & Speedup

Generate (LOWER better) vs audio length (context)



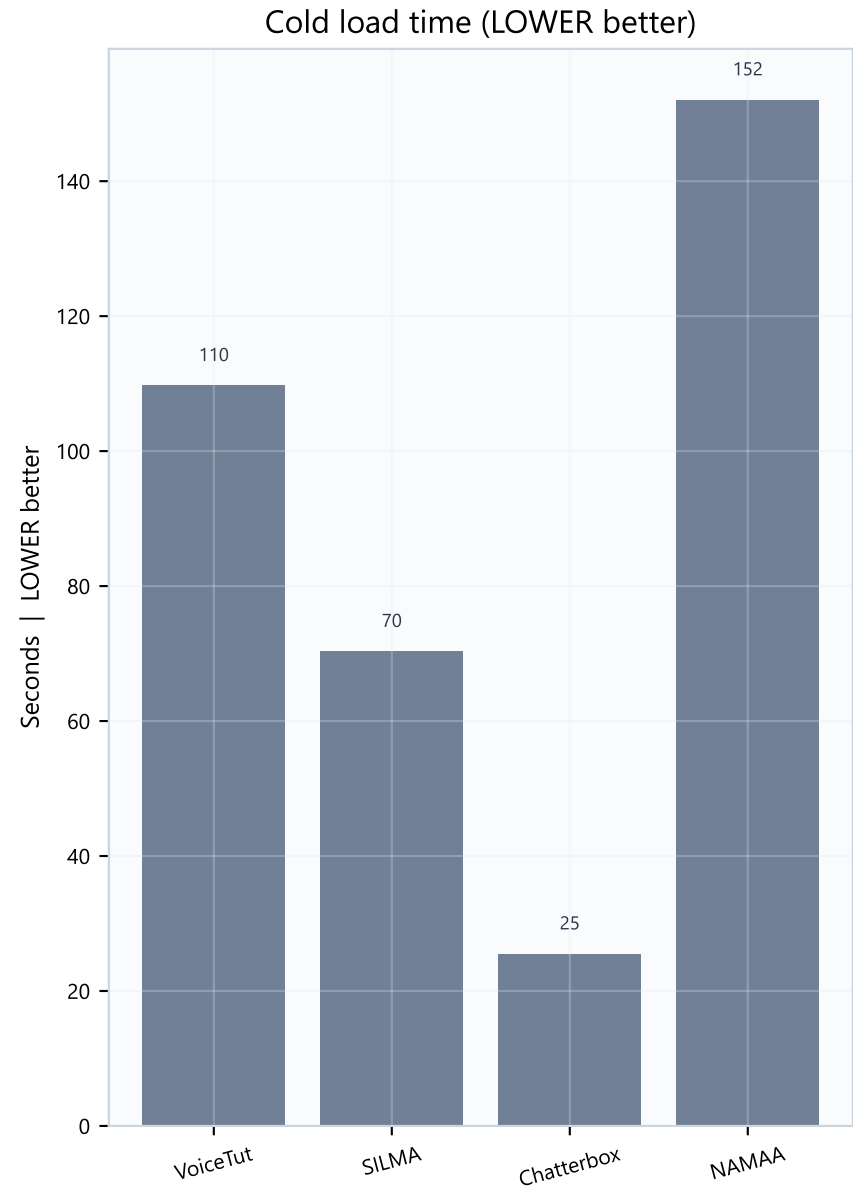
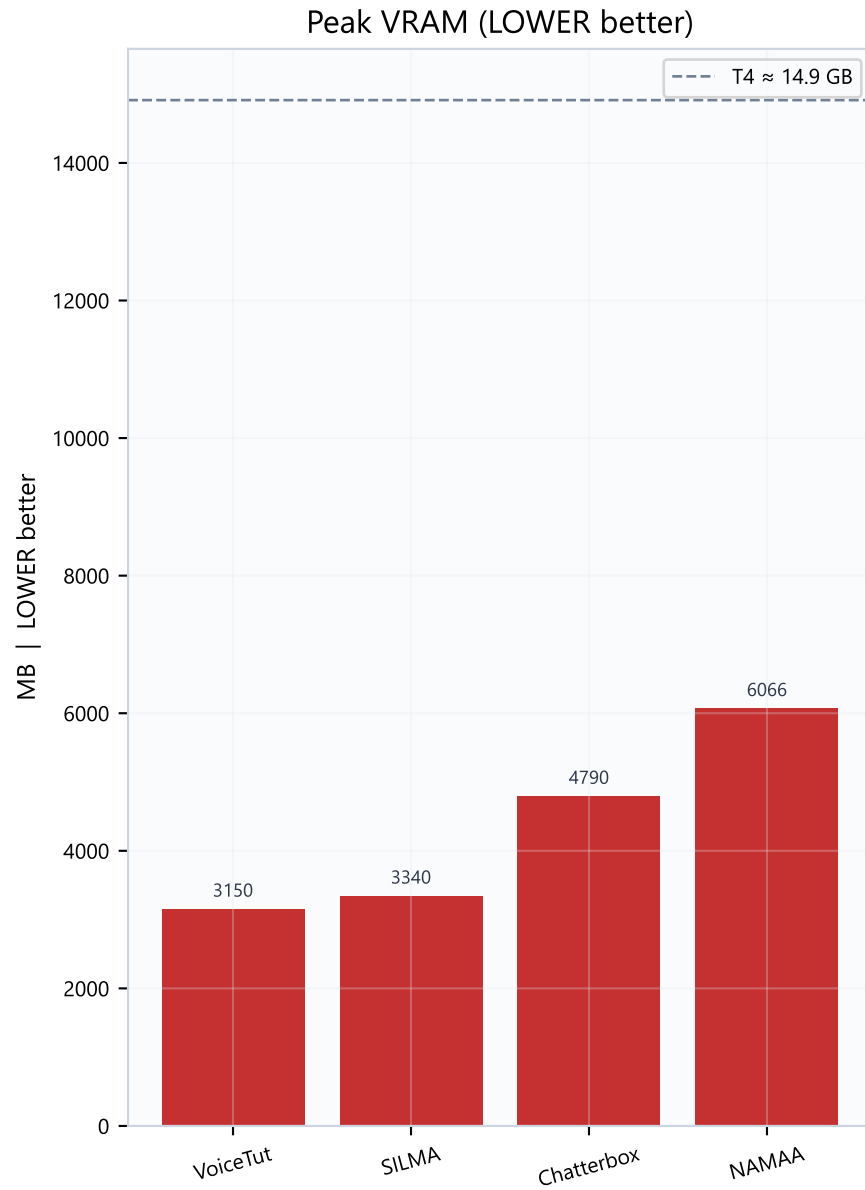
Speedup vs audio duration (HIGHER better)



Reading the timing charts

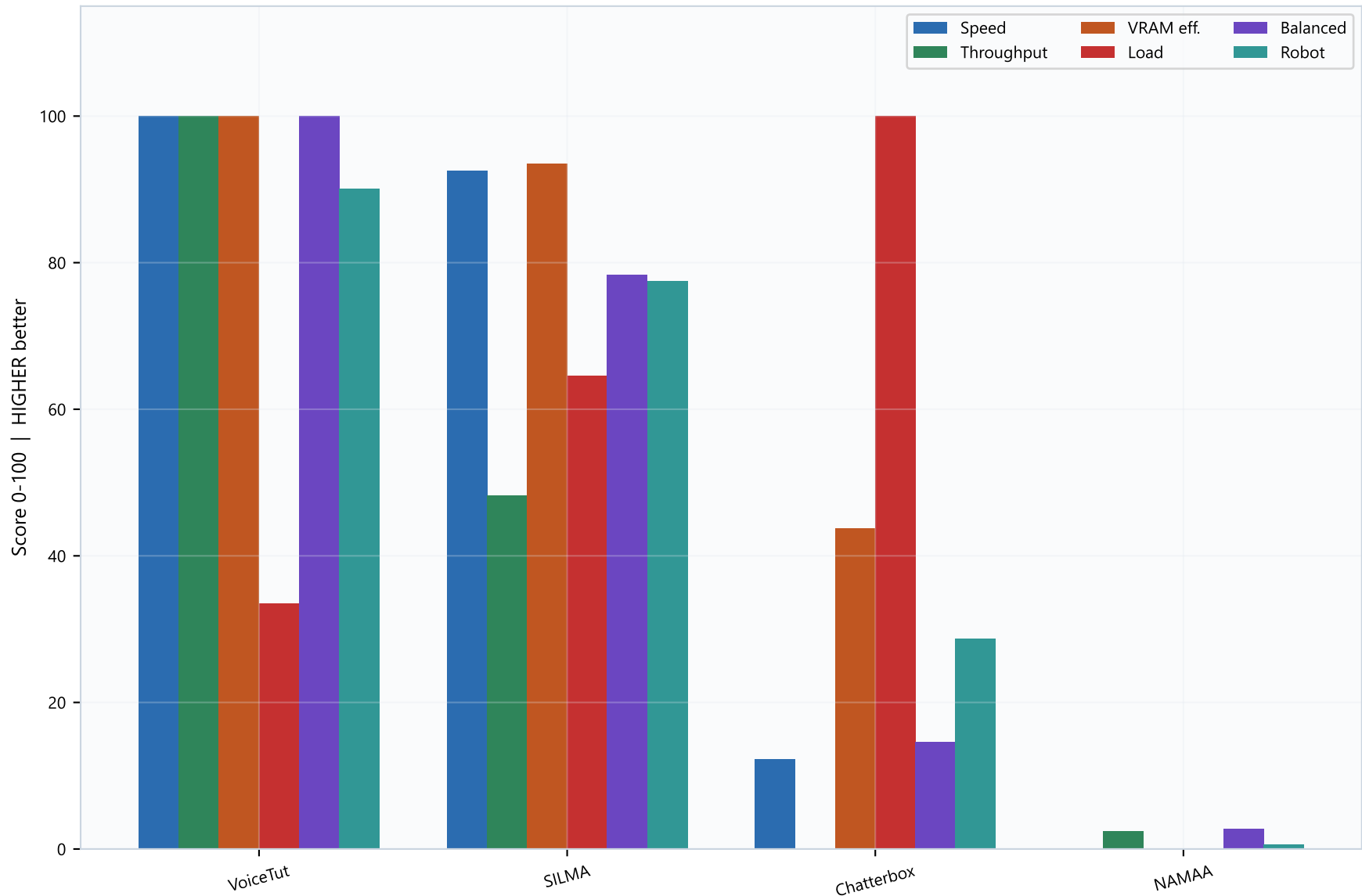
When generate time < audio length, the model is faster than realtime (good).
VoiceTut generates ~131s of audio in ~20s (~6.4x). SILMA ~4.2x.
Chatterbox and NAMAA generate slower than the audio they output.
Cold load time is separate — it hits startup, not every reply, if the model stays warm.

Resources (LOWER better on both charts)



VoiceTut uses the least peak VRAM (~3.15 GB) among OK models — best for co-residency with ASR+LLM. NAMAA is heaviest (~6.1 GB). Chatterbox loads fastest but is not realtime and uses more VRAM.

Normalized score breakdown (all HIGHER better)



VoiceTut sweeps speed/throughput/VRAM/balanced. SILMA is the clear #2. Chatterbox only leads on cold-load score — not enough for live robot use. NAMAA ranks last on robot score.

Listening checklist & how to choose

Listen to these WAVs (score 1–5 each)

- 1) Egyptian dialect naturalness
- 2) Arabic / English code-switching
- 3) Numbers, dates, and times
- 4) Prosody, pauses, and artifacts
- 5) Overall robot-voice suitability

Files: VoiceTut-TTS.wav · SILMA-TTS.wav · Chatterbox-Multilingual-V3.wav · NAMAA-Egyptian-TTS.wav

How to choose

- 1) Default speed/resources pick → VoiceTut-TTS (also listen to confirm voice quality).
- 2) If VoiceTut fails the listen test → try SILMA-TTS next (still realtime, similar VRAM).
- 3) Prefer Chatterbox/NAMAA only if their voice quality is clearly superior AND latency is acceptable.
- 4) Confirm combined VRAM with ASR + LLM on the target GPU.
- 5) For production, also measure streaming time-to-first-audio (not covered in this bake-off).